

Source-Locator Census for the H1 Crosswalk: Native-Key Collisions and the Absence of Production Occurrence Edges in the Frozen Corpus

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Abstract

The H1 branch of the fixed-shift program currently stops at a theorem-backed cut-to-occurrence provenance crosswalk. We turn that informal phrase into an auditable source-locator contract. A positive production edge must name a canonical source path and hash, an exact theorem locator, an exact formula locator, and a nonempty derivation abstract-syntax tree, while preserving the distinction between a formal shadow and an actual occurrence.

For the frozen declared TPC-153/154/155/156/161/162 corpus, the census contains 2988 distinct cut paths, two qualifying positive shadow claims, and two formal-only scoped-obstruction claims. It contains no qualifying actual-occurrence edge in any of thirteen required semantic classes. Independently, the archive has only 866 distinct native triples $(\ell, k, d_{\text{nat}})$. Of these, 854 collide; 2976 rows lie in collision classes, and the excess over native keys is 2122. The exact multiplicity distribution is

$$\{1 : 12, 2 : 220, 4 : 634\}.$$

Consequently the native triple is not a row key for this archive.

Both conclusions are deliberately scoped. Zero qualifying edges in the frozen corpus is not a nonexistence theorem for actual occurrences, and failure of the native triple as an archive key is not a failure of every enriched occurrence representation. The next forced structural task is to identify the smallest archived separating key before attempting source-backed local occurrence patches.

Keywords: fixed shift; occurrence provenance; source locator; archive key; collision obstruction; proof audit.

Claim boundary. This paper proves an exact theorem about a frozen declared source corpus and its archive keys. It does not prove that actual occurrences are absent, construct a production occurrence crosswalk, certify actual active support, select a canonical or minimal parent, establish positive fixed- X L2, give a prime-pair lower bound, or prove the twin-prime conjecture.

1 Why a source census is now necessary

TPC-153 constructs a conservative, row-separated map from every nonsoft cut path to a *partial shadow row* [1]. Its universal property says that any metadata-faithful actual completion must forget back to this shadow; it does not construct such a completion. TPC-154 constructs two inequivalent *formal* completions and thereby stops recovery from current artifacts alone; it explicitly leaves the augmented theorem-backed route open [2]. TPC-155 supplies a typed verifier, but its production witness is absent [3]. TPC-156 therefore identifies

$$\text{H1.theorem_backed_occurrence_provenance_crosswalk} \tag{1.1}$$

as the first missing map object [4]. TPC-161 and TPC-162 preserve this blocker while integrating the arithmetic return side [5, 6].

The phrase “theorem-backed” has two possible failure modes. First, a row may cite a file without identifying the mathematical statement or derivation that creates the edge. Second, a valid shadow theorem may be silently promoted to actual-occurrence semantics. The present paper closes both loopholes by defining the admissible evidence unit before counting it.

2 Frozen corpus and evidence contract

Definition 2.1 (Frozen declared corpus). The corpus $\mathcal{C}_{\text{src}}$ consists of the canonically normalized UTF-8/LF versions of the following eleven production artifacts:

1. the TPC-153 paper, certificate, and 2988-row shadow archive;
2. the TPC-154 paper, certificate, and formal-completion archive;
3. the TPC-155 production status and verifier audit;
4. the TPC-156 H1 route decision;
5. the TPC-161 occurrence–return manifest; and
6. the TPC-162 MVP6 snapshot.

Every source is locked by its repository-relative path and SHA-256 digest. A digest is an integrity assertion only; it does not confer theorem semantics.

Definition 2.2 (Source-locator admissibility). A positive production claim e is *source-locator admissible* if its record contains:

- (i) a canonical path $p(e) \in \mathcal{C}_{\text{src}}$ and its recomputed canonical digest;
- (ii) an exact theorem locator that resolves in $p(e)$;
- (iii) an exact formula locator that resolves in $p(e)$;
- (iv) a nonempty derivation tree $A(e)$ specifying how the cited formula produces the claimed typed object; and
- (v) a semantic class equal to the conclusion of that derivation.

A shadow conclusion therefore cannot inhabit an actual-occurrence class merely by changing a status string.

The derivation tree is intentionally modest. It is not a proof assistant kernel. It records the operation, inputs, key fields, and exact coefficient so that an audit can reject empty provenance and semantic promotion. Full mathematical truth still resides in the cited source.

Definition 2.3 (Actual H1 edge classes). The production census checks the following thirteen classes:

actual occurrence, actual active support, canonical/minimal parent,
 ordered stage, exact multiplier, Q_D , Q_Z , G , P_{h_0} ,
 native affine crosswalk, physical cover, reconnection, occurrence registry.

These are actual-carrier classes. A partial shadow edge is not a member of any of them.

3 Exact source-locator census

Theorem 3.1 (Frozen-corpus edge census). *In $\mathcal{C}_{\text{src}}$:*

- (a) *exactly two shadow claims satisfy [definition 2.2](#): the TPC-153 cut-to-shadow basis injection and its column-conservation identity, both with semantic class*

PROVED_L1_STRUCTURAL_SHADOW_ONLY;

- (b) *exactly two TPC-154 claims satisfy the source-locator contract: formal completion-fibre nonuniqueness and the current-artifacts-only recovery obstruction, both with class*

FORMAL_ONLY/SCOPED_OBSTRUCTION;

- (c) *none of these four claims has actual-occurrence semantics, and the number of source-locator-admissible production edges in every class of [definition 2.3](#) is zero.*

Thus the production actual-occurrence edge family is

$$\mathcal{E}_{\text{occ}}(\mathcal{C}_{\text{src}}) = \emptyset. \quad (3.1)$$

Proof. The audit first recomputes all eleven path-digest locks. It then resolves both LaTeX labels for each positive TPC-153 claim and checks that its derivation tree is nonempty. The first tree is the basis injection

$$e_c \mapsto e_{\text{shadow}(c)} \quad \text{with coefficient 1,} \quad (3.2)$$

and the second applies the column-sum operation to [equation \(3.2\)](#). Their result types remain shadow-only.

The audit next resolves the TPC-154 completion-fibre and obstruction theorems. Their derivations use two conservative formal completions with respectively one unit-weight child and two half-weight children. The source certificate explicitly records that neither completion is an actual-carrier lift and that only the current-artifacts-only recovery route is stopped. Their semantic class therefore remains FORMAL_ONLY/SCOPED_OBSTRUCTION.

Next the audit applies an explicit per-class mapping to the frozen production status, decision, manifest, and snapshot for the thirteen actual classes. TPC-155 records no production witness, and TPC-156/161/162 all retain [equation \(1.1\)](#) as unavailable. No record supplies the required path, digest, two locators, derivation tree, and actual semantic class. Hence every class count is zero. Mutation tests verify that a fabricated edge, hash drift, a missing locator, an empty tree, and a shadow-to-occurrence promotion are each rejected. $\square$

Remark 3.2 (Mapped census, not a future-schema scanner). The thirteen-class table is exhaustive only for the explicitly mapped fields of $\mathcal{C}_{\text{src}}$. The program does not generically discover arbitrary fields added by a future schema. Any corpus or schema extension requires a new mapping and a regenerated certificate.

Remark 3.3 (Closed-world scope). [Equation \(3.1\)](#) is a closed-world statement about $\mathcal{C}_{\text{src}}$. It says

“no qualifying edge is declared in this frozen corpus,”

not

“no actual occurrence edge exists in mathematics.”

The latter would require a completeness theorem for the source universe, which is neither assumed nor proved.

4 Native-key collision obstruction

Each shadow row has a unique source-cut path identifier and carries a native triple

$$\kappa_{\text{nat}}(c) = (\ell(c), k(c), d_{\text{nat}}(c)).$$

One might hope to use κ_{nat} as the future occurrence-row address. The archive itself rules this out.

Theorem 4.1 (Exact native collision census). *For the 2988 production cut paths in $\mathcal{C}_{\text{src}}$, the map*

$$\kappa_{\text{nat}} : \mathcal{C} \longrightarrow \mathbb{Z}^3 \quad (4.1)$$

has 866 distinct values. Its fibre-size distribution is

$$\#\{v : |\kappa_{\text{nat}}^{-1}(v)| = m\} = \begin{cases} 12, & m = 1, \\ 220, & m = 2, \\ 634, & m = 4, \\ 0, & \text{otherwise.} \end{cases} \quad (4.2)$$

Consequently:

$$\begin{aligned} \#\{v : |\kappa_{\text{nat}}^{-1}(v)| > 1\} &= 854, \\ \#\{c : |\kappa_{\text{nat}}^{-1}(\kappa_{\text{nat}}(c))| > 1\} &= 2976, \\ \sum_v (|\kappa_{\text{nat}}^{-1}(v)| - 1) &= 2122. \end{aligned} \quad (4.3)$$

In particular, κ_{nat} is not a row key for the frozen archive.

Proof. Group the 2988 distinct source-cut path identifiers by the exact integer triple (4.1). The resulting multiplicity counter is (4.2). The internal checks

$$12 + 220 + 634 = 866, \quad 12 + 2(220) + 4(634) = 2988$$

verify the number of keys and rows. Removing the 12 singleton rows gives $2988 - 12 = 2976$ rows in collision classes. The excess is

$$220(2 - 1) + 634(4 - 1) = 220 + 1902 = 2122.$$

Since a key must have singleton fibres, the final assertion follows. $\square$

Proposition 4.2 (Explicit fourfold collision). *The native triple $(3, 171, 1)$ has four distinct archived cut paths, distinguished by*

$$(j_L, j_K) \in \{1, 2\} \times \{7, 8\}.$$

Thus the collision in theorem 4.1 is not caused by duplicate row identifiers.

Proof. The four source-cut path identifiers in the production excerpt encode the four displayed pairs and are pairwise distinct, while their native triples agree exactly. The accompanying sample is copied from the production archive and is marked as an archive excerpt, not a synthetic or actual-occurrence witness. $\square$

5 What the two theorems jointly imply

The two obstructions are logically different:

$$\begin{array}{ll} \mathcal{E}_{\text{occ}}(\mathcal{C}_{\text{src}}) = \emptyset & \kappa_{\text{nat}} \text{ is noninjective} \\ \text{evidence absence in a frozen corpus} & \text{exact combinatorial archive fact.} \end{array}$$

The first blocks an evidence-complete production lift. The formal TPC-154 obstruction additionally rules out canonical recovery from the present shadow fields alone while leaving source augmentation open. The second blocks a tempting shortcut for addressing its rows. Neither prevents an enriched construction from succeeding.

Corollary 5.1 (Required enrichment). *Any production crosswalk that represents all 2988 archived cut paths by distinct rows must use an address strictly finer than $(\ell, k, d_{\text{nat}})$, or supply an independently proved quotient theorem identifying every colliding path. No such quotient theorem is present in $\mathcal{C}_{\text{src}}$.*

Proof. This is the dichotomy between retaining distinct elements of a noninjective fibre and identifying them by a proved equivalence relation. The latter is absent by the source census. $\square$

6 Reproducible certificate

The generator reads the upstream JSON/JSONL artifacts, recomputes canonical hashes and all counts, and serializes:

Artifact	Role
Source census	source locks, positive claims, thirteen-class census, and collision counts
Audit	recomputation checks, semantic boundaries, and mutation regressions
Collision excerpt	explicit fourfold production archive collision

Default mode regenerates the artifacts. The `-check` mode recomputes them in memory and demands byte-for-byte equality. The audit performs real mutations: it changes the candidate object and then runs the same validators. It does not merely store expected Boolean values.

7 Next forced object

Before constructing local occurrence patches, the archive needs a lossless addressing system. TPC-163 therefore leaves one immediate finite question:

which smallest subset of existing archived fields separates all 2988 source-cut paths?

This is an exhaustive finite-key problem, not yet an actual occurrence, active-support, or canonical-minimality theorem. Solving it removes ambiguity from the later local-to-global gluing interface without overclaiming arithmetic progress.

8 Conclusion

TPC-163 converts the H1 crosswalk blocker into two precise facts. The frozen declared corpus contains no theorem-backed actual occurrence edge under an explicit source-locator contract, and the native triple fails dramatically as an archive row key. At the same time, the paper preserves the correct open-world boundary: new source-backed occurrence mathematics may still supply the missing edges. The next step is therefore constructive—first refine the archive address, then formulate compatible local occurrence patches.

References

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